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Digital twin representation of socio-technical systems through a distributed co-simulation approach for crisis management

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Abstract

The increasing frequency and severity of significant risks to public safety posed by natural disasters or human-induced events have underscored a critical need for evaluating the vulnerability of urban regions with a focus on their essential infrastructures. This paper presents a novel methodology for the virtual representation of infrastructure vulnerabilities and functional impairments during hazard situations. Thereby, focus is on the mapping of interdependencies among critical infrastructure systems and the cascading effects that can arise from failures within these heterogeneous sectors by means of a digital twin representation. An integration of simulation models for urban infrastructure components, particularly in relation to the built environment and emergency response systems is introduced. Leveraging a modular co-simulation architecture, the framework facilitates the analysis of cascading effects across multiple infrastructure systems, such as water, electricity, gas, and telecommunications. As a proof-of-concept example, urban flooding due to heavy rainfall is considered to illustrate the framework's capabilities in predicting system states and assessing structural impacts on critical infrastructures as well as consequential ramifications for emergency relief units. The findings contribute valuable insights, thereby advertising the utilization of the presented methodology in decision-making and training resources aiding the enhancement of the resilience of urban environments against both natural and intentional threats.

Keywords Digital Urban Twin · Socio-technical systems · Distributed co-simulation · Crisis management

1 Introduction

Evaluating the susceptibility of urban regions and their essential infrastructure to both natural disasters—like floods, intense rainfall, as well as severe storms—and human-induced events—such as terrorist attacks—has become a vital area of research in recent years. This is largely due to the global rise in the frequency and severity of these incidents. Failures or disruptions in critical infrastructure can lead to serious consequences, including ongoing shortages and major threats to public safety. Of particular concern are the hidden interdependencies and unexpected cascading effects among various critical infrastructure systems, which may endanger citizens.

To improve the resilience of urban areas and their critical infrastructure, it is crucial for policymakers, agencies, governments, and infrastructure providers to gain a deep understanding of not only the vulnerabilities and failure risks of individual infrastructures but also the potential impacts

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that the breakdown of one infrastructure can have on others. Previous research efforts have identified risks and hazards associated with cascading effects in interconnected infrastructure systems. The authors of Koliou et al. (2018) summarize the current state of knowledge and challenges related to community resilience and system interdependencies.

1.1 Research gap

Although researchers have documented some interdependencies among critical infrastructure systems, these findings have rarely been applied to integrate simulations of the components within an urban system. Some projects have created tools like SAT (Severity Assessment Tool) (Hayes et al. (2022)) and CReDo (Climate Resilience Demonstrator) (Wang et al. (2022)) to assess and visualize these interdependencies for decision-making purposes. However, specific functionalities of different critical infrastructures and the built environment are typically modeled through data-driven methods rather than engineering approaches, which may fail to recognize the complexity of cascading effects in disaster scenarios. Furthermore, coupled simulations primarily focus on essential infrastructures like water, electricity, and gas networks (e.g., Deshmukh et al. (2011); Praks et al. (2015), while interdependencies with the built environment and emergency response systems are often neglected. In this context, there is a need for simulation frameworks that facilitate the integration and communication among interdisciplinary simulation environments. Previous efforts, such as the Interdependent Networked Community Resilience Modeling Environment (IN-CORE) (Lindt et al. (2023)), mandate the integration of various services, e.g., including analyses of buildings with fragility curves and simulations of infrastructure networks under various hazard scenarios. However, the methods used for simulation are not discussed comprehensively, and emergency response systems, which are critical for effective training, are omitted.

Regarding the efficient realization of interdisciplinary co-simulations, Restel (2023) introduces the concept of *SimulationOps* that proposes a collaboration process (from creating and combining simulation components to deploying, configuring, and running simulations). This process utilizes the concept of distributed "as-a-service" platforms to achieve its objectives. Numerous co-simulation models and platforms have been proposed. The most widely recognized concept is the High Level Architecture (HLA), publicly available as a set of standards (IEEE (2010)). This concept describes a federated system of multiple simulators, along with a common runtime infrastructure to facilitate communication and synchronization. One example of a simulation platform that uses the HLA approach in the domain of mobility and communication is Eclipse Mosaic (Schrab et al. (2023)). HLA-based systems synchronize the steps for a simulation in a global fashion but lack something like an event stream approach allowing for each service to execute its next step independently if the necessary input is available, which can boost runtime performance. The Functional Mock-up Interface (FMI) (Modelica Association (2025)) is a standard for exchanging models for tightly coupled co-simulations based on XML and the C programming language. Maestro (Thule et al. (2019)) is one example utilizing FMI for co-simulating cyber-physical systems. However, FMI does not provide the development and deployment flexibility to support the iterative development, deployment, and usage process. The *Cybersecurity & Information Systems Information Analysis Center*'s approach to "Modeling & Simulation as a Service" (MSaaS) (Siegfried et al. (2018)) emphasizes the use of cloud computing and service-oriented architectures to deliver modeling and simulation capabilities effectively. "As-a-service" refers to a self-service approach, where users can access and use services on-demand. Gütlein and Djanatliev (2020) presents an architecture for MSaaS, which focuses on Apache Kafka for communication between and coordination of components participating in a simulation. In summary, the existing "simulation-as-a-service" approaches already address distribution, on-demand usage, or using diverse programming languages, but do not support the event streaming approach envisioned here.

1.2 Innovation

To fill these gaps, a new methodology has been established to enable the virtual representation of infrastructure vulnerability and functional impairments in urban districts during hazard situations. It is built upon the concepts presented in Martini et al. (2024). Its modular architecture and integration with the Fraunhofer Open Source SensorThings API Server (FROST) (Hertweck et al. (2019)) make it adaptable and capable of being expanded to include additional hazard scenarios and infrastructure components. This versatility enables various applications, such as decision-support tools for authorities and infrastructure operators, as well as training resources in public safety.

The framework integrates multiple simulation modules for different critical infrastructures and services, including water, electricity, gas, and telecommunications to form a comprehensive digital twin representation of key infrastructures in urban areas. Each module is built on engineering models that are based on physical principles, standards, or (georeferenced) real-world data, allowing for reliable predictions about the modeled system's state under certain conditions. Analyses of scenario-specific building damage using probabilistic engineering models as well as an agent-based simulation of emergency operations, modeling the rapid responses of emergency units to critical incidents are also integrated in the framework. This combination allows

for a structural assessment of critical infrastructure and an exploration of cascading effects that could hinder emergency response efforts.

The co-simulation platform emphasizes a strict separation of the various development and operation responsibilities as well as an extensive flexibility in forms of deployment. It furthermore facilitates an event streaming approach to orchestrate the services involved in running a simulation.

To illustrate the performance of the presented framework a scenario of urban flooding resulting from heavy rainfall, addressing aspects of gas, water, power supply, communication networks, emergency services, and building structures is considered as an example application to demonstrate how the developed methodology can provide valuable insights before, during, and after disasters, enhancing the understanding of cascading effects and chain reactions triggered by significant impacts on critical infrastructure from severe natural events or intentional attacks.

This article is structured as follows. The co-simulation framework enabling the coupled simulation of several critical infrastructures is introduced in section 2. The construction of a digital twin representation of an urban environment by means of the distributed co-simulation framework is discussed in section 3. The simulation modules needed for the modeling of the respective individual infrastructures and services are described in subsection 3.1, while the implementation of their interfaces is detailed in subsection 3.2. As a proof-of-concept of the newly introduced framework, a first use-case example of cascading infrastructure failures in urban environments is considered and the results of this example application are showcased in subsection 3.3. Section 4 comprises the conclusion and highlights future outlook and prospects of the presented methodology.

2 Co-simulation framework

The following section presents the design of a distributed co-simulation framework that has been developed to facilitate interdisciplinary research and development.

2.1 Design objectives

The primary objective of the development effort is to establish a framework which enables time-based co-simulation in a distributed environment. The framework is designed to support the development and utilization of simulation components by an interdisciplinary team of researchers and developers. Integrating existing simulator systems, each with its own user interfaces, simulation control, and communication facilities poses a major challenge.

The framework is designed to be scalable and flexible, possessing the following key characteristics: (i) simplicity of use and extensibility, (ii) capability for the development of complex simulation scenarios, and (iii) support for the execution of such scenarios on local machines, cloud-based infrastructures, or hybrid environments.

The initial considerations have been based on existing approaches such as serverless (stateful functions), event-driven, and service-based architectures. A central objective is to completely separate application and platform functionality. Core data processing functionality is to be separated from the following aspects.

- *Control flow*: simulators are required to surrender, for instance, time control facilities, so that they could be controlled from the outside.
- *Data flow*: simulators are to relinquish control of data exchange to facilitate their seamless integration into flexible data processing pipelines.
- *Communication technology*: a simulator's algorithmic component needs to be separated from any integrated communication mechanisms, to facilitate its deployment in diverse runtime environments.
- *Deployment specifics*: a simulator's functionality needs to enable cross-platform execution, allowing it to run on various environments, including local machines, cloud services, and hybrid configurations.

2.2 Framework components

The framework comprises four major types of components: (i) processing units, (ii) services, (iii) applications, and (iv) orchestrator(s).

Processing units implement simulation algorithms or other data processing tasks as passive and independent building blocks providing a common Application Programming Interface (API) to facility controlling their execution from the outside. Processing units can be written in diverse programming languages.

Services encapsulate processing units and provide life cycle management and inter-process communication interfaces. They offer mechanisms for creating unit instances, initializing them for a simulation run, and invoking their functions to apply external data, execute subsequent steps, or query their state. This enables running multiple simulation algorithms in a coordinated manner in distributed environments and allows processing units to participate in multiple concurrent simulations for various users, objectives, and parameter sets.

Applications describe simulation scenarios. In a typical co-simulation scenario, services are interdependent, i.e., the output of one service serves as the input of another. Figure 1 illustrates this principle in form of a block diagram based on Fundamental Modeling Concepts (FMC) (Knöpfel et al. (2006)) using an example of four

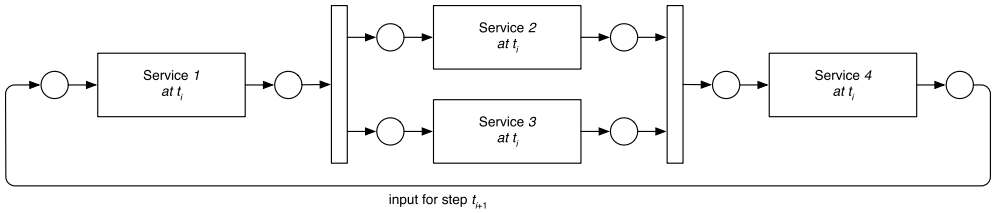


Fig. 1 Service dependencies in a co-simulation environment. The labeled boxes represent the state of services at a time t; information flows along the arrows through channels (represented as circles); the unlabeled boxes represent synchronization points, to illustrate that a service waits for all input to be available before proceeding with its own next step

interdependent services collectively executing the steps of a simulation run. Applications specify the required simulation services, their dependencies, initial state configurations, and simulation parameters, including model time, simulation duration, step widths, and other relevant settings.

Orchestrators accept application descriptions and initiate and execute corresponding simulation runs. Consequently, an orchestrator establishes connections to the requisite services and invokes them sequentially (until the application's termination condition is met), while ensuring that data necessary for a service to execute a specific step are retrieved and provided in a timely manner.

Given that the present research primarily focuses on connecting simulators from diverse domains, the following sections concentrate on how the concept of processing units can be used to implement the necessary functionalities.

2.3 Processing units

Processing units are designed to implement simulation algorithms as self-contained, passive components that are independent of a specific execution environment. At its core, a unit is essentially a function $f(\text{state}_{t_{i-k}}, t_i) \mapsto \text{state}_{t_i}$ (step function) that takes a model state and a corresponding point in model time and produces a subsequent state. There is a well-defined interface for initializing and retrieving model data and for applying external events to a model state, enabling the units to be used in a cooperative environment. A processing unit instance manages its own local model independently of any other unit. It may use external resources such as databases to store and retrieve its model data. Units may also use shared *read-only* resources such as lookup services. Units can be written in any programming language that has a language binding to the framework; currently this includes JavaScript, Python, Java, Kotlin, and C#.

A processing unit implements the following API.

- `init (options)`: initializes the state of the unit with the given options.
- `apply (time, data)`: applies the given external data to the state of this unit for the given point in model time.
- `step (time)`: computes state changes for the given point in model time.
- `query (time, filter)`: queries the state for the given point in model time using a unit-specific query/filter expression.
- `release ()`: releases all resources held by this unit.

An API contract exists that stipulates additional conditions under which a unit's functions are executed. For instance, the `init` function is called at most once before any other function; the `apply` function may be called between 0 and n times per time step directly before the corresponding `step` function; the `step` function is called exactly once for a time step; the `query` function is called at most once per pair (time/filter); and the `release` function is executed at most once after any other API call.

Depending on data flow patterns, processing units can be classified into three types.

- *Generators* produce data, they, however, do not accept external events. Examples include clocks, replay, interpolator, or control units.
- *Collectors* accept external events, however, do not produce any data. Examples include consoles, archives, displays, or monitors.
- *Transformers* do both, they accept external events and produce data. Examples include simulators, filters, and data converters.

2.4 Unit interaction

Processing units are self-contained, that is, they have the freedom to model and represent data internally without external constraints. However, to facilitate data exchange with other units, a common data model must be established or a mechanism has to be provided for converting data between different models.

Additionally, modeling and communicating metadata (e.g., data source identification, valid time, transaction time) is crucial for facilitating rejection or validation of data records being applied. To address the latter issue generically, the framework encourages the use of the CloudEvent model (CloudEvents Authors (2024)) to encapsulate data for exchange between simulators. CloudEvents offer a standardized metadata structure and well-defined methods for encoding data for a variety of formats and communication protocols.

The remaining challenge of managing different data models can be resolved by employing converter units. When two units are using a common model, data can be exchanged directly, as demonstrated in the following program fragment.

```
...
data := source.query(t);
target.apply(t, data);
...
```

In most cases, particularly when simulators cover diverse domains or have been developed by distinct teams, data conversion is indispensable. Data converters are specialized transformer units that accept data from one model and convert it to a target model. The following fragment illustrates the flow of data between simulators in a schematic manner.

```
...
data_src := source.query(t);
transformer.apply(t, data_src);
if exists(transformer.step)
    transformer.step(t);
data_tgt := transformer.query(t);
target.apply(t, data_tgt);
...
```

The descriptions thus far have focused on the interaction between processing units, i.e., on communication inside a simulator network. To effectively use the outcome of a simulation, however, a simulator network requires mechanisms for communication with external systems. This is achieved through the use of peripheral units, such as generators or collectors.

Figure 2 depicts an example simulator network that communicates with an external user interface system. The buttons of the User Interface (UI) are used to pause or resume the simulation through a clock control unit, which subsequently generates clock events. Simulator 1 depends on the clock ticks and the results of simulator 3; simulators 2 and 3, in turn, depend on the results of simulator 1. An additional converter unit is employed to convert data between simulator units 1 and 2. The simulation results of units 2 and 3 are collected to be displayed in the UI.

3 Digital twin representation of an urban environment with coupled interdependent infrastructures

To demonstrate the capabilities of the developed framework, the ramifications of a heavy rainfall event in an urban environment and subsequent consequences on critical infrastructures are to be studied as an example application in

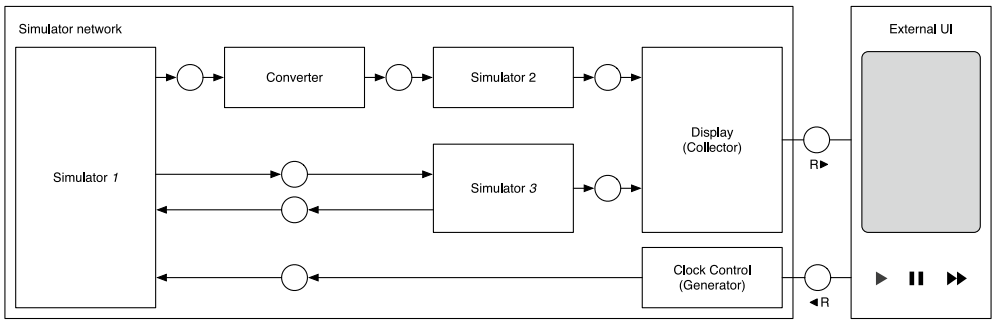


Fig. 2 Schematic FMC representation of units interacting in and outside a simulator network

subsection 3.3. The scenario under consideration takes place in an urban area of a large German city, which is primarily characterized by residential buildings. In addition, there are schools, an ambulance station, a hospital, and two power substations in this area. To model those infrastructures together with their mutual dependencies, a digital twin representation of the urban environment is constructed with a focus on interconnected infrastructure models through the presented co-simulation framework. Figure 3 depicts a possible exemplary configuration: Reconnaissance modules like audibility and visibility analyses, information from spontaneous volunteers or drones are incorporated as generator units providing the input for the coupled simulation. Relevant interdependent critical infrastructures are represented by coupled simulator units which can process the received input and can interact by means of the co-simulation framework. The outputs of those coupled simulations are gathered by dedicated collector units for visualization and analyses (see Winter et al. (2024)). The implementation and modeling of the respective units constituting the digital twin representation are detailed in subsection 3.1. The realization of interactions among the individual simulators are outlined in subsection 3.2.

3.1 Model overview

Digital models with simulation capabilities of the distinct components of infrastructure networks present in the considered urban area—namely gas, power supply, drinking water, and telecommunications—alongside emergency response and building vulnerability, constitute the building blocks of a digital twin representation of critical infrastructure in urban contexts for evaluating its interdependent functionality and resilience. Consequently, a dedicated simulation tool is required for each component, tailored to fulfill its specific requirements and operational functions. Here, modeling of supply infrastructure operations and assessment of the building vulnerability is based on physical principles like mechanics, thermodynamics, electrodynamics, and hydrodynamics as well as derived engineering models. For the dynamic simulation of the emergency response, an agent-based modeling approach is employed. Reconnaissance is exemplified by input gathered through spontaneous volunteers, while a suitable visualization dashboard is introduced. Each tool operates autonomously, serving as an independent simulation instrument, which will be detailed in the subsequent sections for all components. This subsection also serves as a demonstration of the framework's capability of and flexibility in incorporating and connecting versatile, heterogeneous, interdisciplinary building blocks, based on specifically developed or existing tools. Hence, operational and implementational details, underlying theoretical footing as well as required embedding and inputs together with provided outputs are summarized for each individual unit to be employed in this exemplary multilateral modeling of the urban configuration in question. In particular, focus is placed on similarities in physically grounded modeling of supply infrastructure systems (e.g., mapping of gas, power, and water network elements to edges and nodes of graph-based descriptions) as well as conceptually different approaches of, e.g., agent-based modeling of response unit dynamics. The development of the individual building blocks might require considerable amounts of effort—depending on their respective degree of novelty. The following subsections cover their most relevant aspects suitable to underline the aforementioned deliberations, whereas the building blocks based on newly developed tools are introduced in a bit more detail.

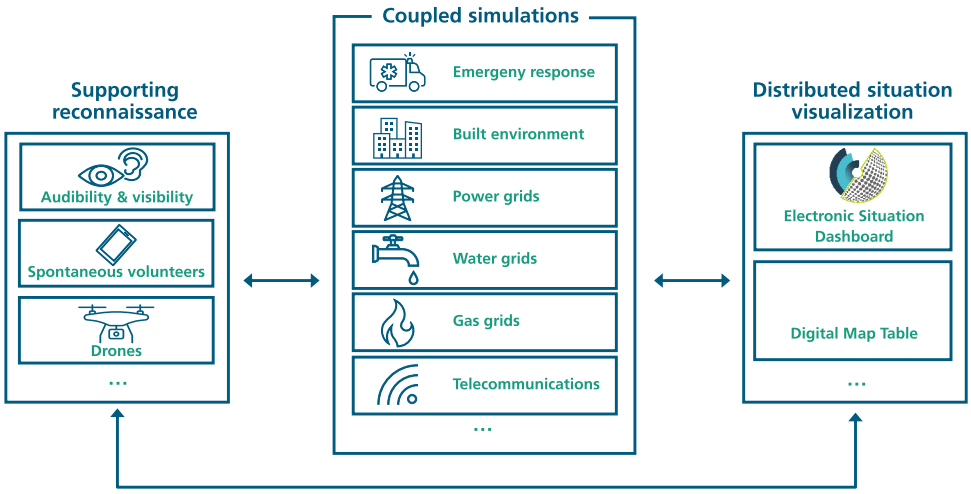


Fig. 3 Coupled simulations example

3.1.1 Simulation of extreme contingencies in urban gas distribution networks

Numerical simulation of natural gas grid operation can provide useful aid for decision support in economic as well as safety considerations. Within current energy sectors, natural gas infrastructures still play a prominent role while offering a potential vantage point for hydrogen expansion, e.g., by step-wise hydrogen blending. Concomitant with its importance, the safety of gas supply has particularly been shifted into focus due to acts of sabotage as well as political sanctions.

The hydraulic gas-network simulation module employed in the presented co-simulation framework is developed as a Python module predicated on Ganter et al. (2023) as follows.

Natural gas grids can be represented as networks comprising nodes—indicative of sources, consumers, junctions, etc.—that are interconnected by directed, one-dimensional edges symbolizing pipeline segments, compressors, regulators, and valves. The resulting network graph, enriched with operational parameters such as source pressures, consumer demands, and the characteristics of compressors, regulators, and valves, as well as pipeline attributes including lengths, diameters, inclinations, and roughness values, constitutes the input for the gas module. The dynamics governing natural gas, a mixture of gaseous hydrocarbons, are dictated by the principles of thermodynamics and hydrodynamics, ensuring the conservation of mass and momentum under the influences of inertia, friction, gravity, and external forces from components such as compressors (Ganter et al. (2023)). Applying these principles to each pipeline section and each node, a system of coupled nonlinear algebraic equations that delineate the system state in terms of the average flow rate through each section and the pressure at each node can be derived (cf. Ganter et al. (2023)).

Upon receiving an attributed gas-network graph as input, the gas module automatically constructs the corresponding system of nonlinear algebraic equations based on the topology and attributes of the input graph. To ensure mathematical solvability, the viability of the input graph is scrutinized, and the flow-regulator concept introduced by Ganter et al. (2023) is employed to facilitate physical solutions that may diverge significantly from design-point operations. For enhanced robustness, the solution is derived through an iterative linearization process combined with a least-squares optimization method.

The incorporation of the gas-network simulation module into the co-simulation framework builds on the deliberations presented in Martini et al. (2024). The module is implemented as a simulator unit transforming gas-network-graph data to corresponding numerical values for the hydraulic quantities describing the flow characteristics within the network (i.e., flow rates and pressures). Hence, specific data converter units are required when connecting the gas-network simulation module to other interdependent simulation modules in order to, e.g., interpret received data on power supply in terms of the operational state attributes of power-dependent gas-network elements or to map the resulting numerical values of the gas-flow characteristics to the supply status of gas consumers.

3.1.2 Simulation of urban power distribution grids

The simulation of electrical power grids requires granular modeling of their network components to detect critical infrastructure contingencies due to damaged power assets or insufficient power supply. In the context of power system analysis, the network is modeled as nodes (bus bars) that host generators and consumers, along with edges (branches) that represent lines and transformers. To determine voltage amplitudes and angles, from which the power flows and system losses can be derived, this configuration can be expressed as a mathematical problem known as the load flow problem. In this context, power flow equations define the balance between power injections and consumptions at each bus bar. In addition to the interdependencies of the individual elements, other relevant parameters, such as the impedance and length of the lines and the nominal power of transformers and generators, are incorporated into these equations (cf. Grainger et al. (2016)).

The variety of analysis and simulation tools for power systems capable of solving the load flow problem is reviewed in Martini et al. (2024). The presented power grid module uses Pandapower (Thurner et al. (2018)) for its easy integration into existing Python code and performance in complex networks (Grishin et al. (2024)). In addition, Pandapower has a standard type library for common elements of the power grid with standard type parameters.

The power grid module either uses a Pandapower network model as initial input or generates one based on geodata. To create a computable network model, the geodata must meet specific requirements: (i) The data must represent a coherent network model in which all nodes and edges are connected. (ii) Consumption must be covered either by generators or external networks. (iii) All type parameters of the elements, such as nominal power, resistance, and reactance, must be clearly defined. (iv) There must be a defined slack bus bar that serves as a reference point for the voltage and power balance.

The power system is simulated in an event-oriented way, allowing for the representation of load changes and grid topology adjustments. After each modification, a load flow calculation is performed with a subsequent check for component overloads and voltage band violations, adjusting the network as needed until all boundary conditions are met.

To interact with the co-simulation framework, data converters provide the required data that imply modifications to the power network. This input data can include event information on damaged and repaired infrastructural assets, along with generator and load scaling changes. For outputs, data can be queried either for damaged power assets or the power consumption of infrastructural assets. Various data converters are utilized for translating information to the stack of infrastructure components that depend on the power system.

3.1.3 Simulation of drinking water networks

Drinking water distribution networks are complex systems composed of interconnected components. These networks can be represented as directed graphs, where the nodes correspond to infrastructure elements, such as junctions, reservoirs, and tanks, while the links represent pipeline sections, pumps, and valves that connect the nodes. Each network component has specific parameters, such as flow rates, pressure levels, and diameters, which are essential for simulating the system's behavior. Hydraulic simulations are used to accurately model and optimize the performance of these networks. The goal of these simulations is to model water flow and calculate key parameters, like pressure, velocity, and flow rate at different points. These simulations are based on fluid dynamics principles, where the flow through each pipe follows the laws of conservation of mass and energy.

Sophisticated computational models are employed to simulate hydraulic conditions, with one of the most widely used tools being EPANET (Rossman et al. (2020)). EPANET is an open-source software that models both the hydraulic behavior and water quality dynamics of drinking water distribution systems. It solves nonlinear algebraic equations based on flow equations and applies hydraulic principles to simulate the steady-state and dynamic behavior of the network.

EPANET serves as the computational engine for various water network simulation platforms. Tools like WNTR (Klise et al. (2017)) and QGISRed (Martínez Alzamora et al. (2019)) extend EPANET's capabilities; however, they come with limitations, such as the lack of usability for non-programmers or the requirement to operate within specific environments (Martini et al. (2024)). HydroDyn Map was developed to overcome these challenges, combining EPANET's computational strength with .NET and C# technologies to provide a more accessible and user-friendly interface. The tool also integrates spatial data by enabling the assignment of geographic coordinates to EPANET elements within a specific Spatial Reference System (SRS) (Chang (2018)). This integration enhances simulation accuracy by incorporating real-world geographical data, enabling advanced spatial analyses, such as including population density in the simulation to assess the system's efficiency in serving different areas. Additionally, HydroDyn Map supports the storage of network data and parameters in a PostGIS database (PSC & OSGeo (2023)), making it easier to manage, query, and integrate with other data sources. By bridging hydraulic simulations with spatial data, this tool provides a more seamless and comprehensive approach to modeling drinking water distribution networks.

Although initially developed for Windows, the software can be adapted to run on any platform compatible with the .NET framework. By offering the option to deploy HydroDyn Map within a Docker container, it ensures complete independence from the user's host architecture. Furthermore, the software wrapper supports a range of standard interfaces, enhancing interoperability and ease of integration.

The water module in the co-simulation framework utilizes HydroDyn Map as a tool to simulate drinking water networks. The simulation module can receive information about the operational state of network elements and send parameters, such as head, pressure, demand, and demand deficit, as output. In order to connect with other modules, data converter units are required to enable the water module to interpret modifications to its network introduced by other modules and to return the updated state of the drinking water network.

3.1.4 Simulation of (LTE/5 G) cellular network coverage in disaster-stricken areas

The cellular network communication module aims to simulate cellular communication outage based on critical infrastructure failure. Cellular networks are a fundamental component of modern urban infrastructure, serving critical roles for individuals, essential services, and public systems. Citizens depend on cellular networks (in the absence of Wi-Fi) for receiving warnings, initiating emergency calls, and accessing crucial disaster-related information. Critical infrastructure systems, such as distributed electricity or water networks as mentioned before, rely on these networks for remote operations and management, ensuring continuous functionality even during disruptions. Critical infrastructure operators face significant disruptions in the absence of reliable cellular communication, as they rely on it to coordinate field staff and ensure seamless operations.

Network simulation is a commonly known methodology that is adopted not only in research but also widely in the private/public sector. It is therefore beneficial for novel co-simulation approaches to include and integrate scientifically validated approaches and tools. This includes, next to scientifically evaluated network-validation approaches (Wrulich et al. (2008)), the real-world impact on network models and protocols, such as discussed in Jardosh et al. (2005) and Hahn et al. (2015). Notably, the simulation of cellular networks has been variously explored in prior research, encompassing link-level and system-level analyses in Rupp

et al. (2016) or Virdis et al. (2014) based on OMNeT++, as well as system-level approaches as presented in Ikuno et al. (2010) and Johnson (1999) utilizing MATLAB. However, all of these methods are not bound to one specific tool but are rather tool-agnostic. Further, coupling various simulation tools in an integrated manner to investigate communication protocols, networks, and network coverage is not a novel approach and has been a successful simulation approach, such as in V-to-X related research (Protzmann et al. (2017)), next to other research domains. The here-presented network simulation module calculates the effects on cellular network coverage in the target area that arise from critical infrastructure failure, such as a large-scale power outage, cable destruction, or destruction of buildings. Based on this input, the module generates geographical forecasts that resemble the remaining coverage within the disaster-stricken area and additionally evaluates the citizens' remaining ability to either receive public alerts and warnings or initiate emergency calls (Lukau et al. (2023)) using their cell phones.

The simulation module itself is implemented in a system-of-systems architecture with multiple integrated subsystems, databases, and submodules bound to specific tasks. The submodule that is bound to network calculations is the core element of the module and includes a code-level variation of the NS-3-based discrete-event simulation framework (Riley and Henderson (2010)). This submodule recreates the network topology based on real-world infrastructure data that contains the positions and configurations of each mobile base station in the target area. The source-code variation of the NS-3 framework enables the dynamic integration, load-balancing, and scalability as well as parallelization of large scenarios. It further enhances each simulation run with a specific simulation state. This provides the ability to dynamically simulate various scenarios and changing network topologies of the public cellular network infrastructure and also enables dynamic establishment of the network scenarios and signal processing calculations based on changing scenario requests. The LTE architecture modules and signaling models as well as functionalities that are used in the NS-3 variation have been introduced and validated in Giuseppe et al. (2012) and Baldo et al. (2011).

Integrated into the co-simulation framework, the module is implemented as a processing unit with its subservices and submodules as self-contained and masked components. It is initialized with an internal database of buildings, base stations, and their configurations that equally resemble their real-world position and configuration in the simulated scenario. The simulated base stations are then configured with the respective height (with respect to building height), transmission power, frequency bands, bandwidth, and corresponding frequency channels. In a running co-simulation (i.e., the simulation of a blackout in the target area), the processing unit internally monitors the power supply state at each base station. Upon receiving new power supply information the current internal simulation state is applied, altering functioning/non-functioning base stations. Since the base stations atop the buildings are supplied by the building's own power supply system, the power outage directly influences the antennas on the roof. As this information originates from different processing units (i.e., simulation of power distribution grids), it may have been altered by an external co-simulation framework converter that translates simulation results from electricity-related to communication-related domains. The applied topological change in the network is recalculated by a single internal simulation run, resulting in an updated coverage area as a new simulation result. To determine the new coverage area or size of each cell per base station, the uplink and downlink between simulated user equipment (i.e., smartphones) and functioning/non-functioning base stations in the area is simulated. The calculated Received Signal Strength Indication (RSSI) measured at each user equipment, as well as the simulated cell-selection process of each user equipment, define the geographical coverage area in which users are predicted to either have service or not. The simulation results are served back into the co-simulation framework toward coupled processing units through dedicated data converter units.

3.1.5 Simulation of the emergency response

In the presented framework, most of the components describe different technical systems that are quite entangled with one another, but there is another important aspect when modeling failures in the critical infrastructure in order to understand their impact on society and daily life. While some failures are fixed automatically or can be resolved remotely by the operators, other failures must be resolved by humans, either because of their complexity or quantity. Incorporating this human interaction into the co-simulation framework allows for a more comprehensive description of the socio-technical systems.

Human interaction involves repairing and maintaining parts of the critical infrastructure and affiliated machines as well as the work of authorities and organizations responsible for public safety. Their simulation is typically achieved with Discrete-Event Simulations (DES) or Agent-Based Simulations (ABM).

Discrete-event simulations examine the behavior of complex systems by modeling a set of events (e.g., failures, emergency calls) that interact with the system at specified time points. These simulations operate along a timeline of discrete time steps, generating a system state for each time step (Mahdiraji et al. (2022)). Agent-based simulations focus on replicating individual behaviors through parameterized agents (e.g., people, vehicles) that act as state machines by perceiving, processing, and reacting to

their environment (Andrae and Pobuda (2021)). This agent-level perspective enables a detailed analysis of microscopic processes in a complex system, such as compliance with response time targets (Bandyopadhyay and Singh (2016)).

There are various simulation tools available for this task: The simulation software MATSim (MATSim Contributors & MATSim Association (2025)) is a multi-agent simulation framework designed for traffic simulation in urban areas. It can be used to estimate emergency response times based on underlying traffic conditions (Koch et al. (2020)). Similarly, SUMO (Alvarez Lopez et al. (2018)) is a traffic simulation tool that can be applied to simulate traffic incidents and emergency service vehicles (Li et al. (2024)). Other agent-based simulation frameworks are provided with AnyLogic, NetLogo, and Repast (The AnyLogic Company (2025); Wilensky (1999); Argonne National Laboratory (2022)). Anylogic is a framework which allows to replicate the procedural steps of systems, for example the patient flow in an emergency department (Murtaza Sara et al. (2024)). NetLogo provides agent-based simulations as a black box, in which a patient flow can be modeled through an emergency department (Godfrey et al. (2023)). Repast Suite enables the agent interaction in a domain-specific point of view to reproduce the behavior in certain scenarios (Cabrera et al. (2012)).

The emergency response simulation was developed as a discrete-event simulation to handle incidents with unknown entry time combined with an agent-based simulation approach for dynamic representations of first responders, vehicles, detectors, etc., and a domain-specific agent definition of a control center. It is implemented in the Kotlin programming language in a domain-driven design (figure 4). Incidents stand for failures, accidents, or other events. They are detected by some kind of sensors or humans. These detectors report the incident to a control center at which a number of tactical units are dispatched to resolve the incident.

Incidents can be generated within the emergency response simulation or are received from other modules of the co-simulation framework. The detectors obey different policies and logic ranging from simple automated alarms to humans calling the emergency hotline. When the incidents are reported to the control center, there is a different set of policies and logic dealing with them and dispatching available units on operations to resolve the incidents. All simulation logic must be adapted to the designated modeled area and the applied policies.

The emergency response simulation is a processing unit as described above. It receives inputs in the form of cloud events with information about new incidents, namely their properties and locations. The output comprises updated information of its entities: incidents, operations, tactical units, and depots.

Exemplary, in the application in section 3.3, a heavy storm leads to structural problems of buildings and critical infrastructure, buildings get flooded, streets and bridges are impassable, and the electricity and communication network fails in certain areas. Additionally, there are weather-unrelated incidents which emerge within the emergency response simulation based on calibration data of the modeled area. The incidents due to the heavy storm are received from external co-simulation modules. For example, flooded buildings and impassable streets trigger a set of police, fire, and medical emergency units. These tactical units drive to the incidents based on current road conditions and possible road blocks, then they work on the incidents with their individual policies and logic. When the incidents are resolved, the units drive back, bring injured people to hospitals or get assigned to new operations.

The emergency response simulation allows studying several different aspects in a scenario. It can vary by

- different initial conditions, e.g., time of day and spatio-temporal availability and distribution of resources and forces,
- different policies and logic of the simulation agents (detectors, control center, units), e.g., from fixed processes, control theory, or reinforcement learning,
- different inputs from the other co-simulation modules.

Studying simulations under varying conditions enables a better understanding and forecasting of the crisis situation. It also helps to evaluate and improve disaster management plans, distribution of resources and staff planning, and can be used in training environments for staff in operational headquarters.

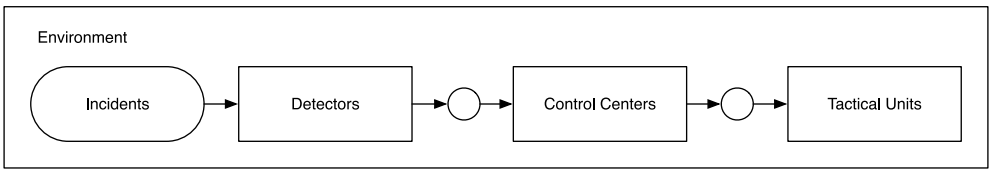


Fig. 4 Implementation concept of the emergency response simulation with the main entities incidents, detectors, control centers, and tactical units

3.1.6 Hazard vulnerability assessment of the built environment

Flood hazard assessment of buildings and structures comprises various methods and approaches. A comprehensive overview that summarizes the current state of the art and provides detailed information about the methods is presented in Martini et al. (2024); Vetter et al. (2024); Neuhaeuser et al. (2024); Rosin et al. (2024).

In the present research, an approach is developed that enables the site- and building-specific quantification of flood damages and building vulnerabilities. This is accomplished using categorizable, generalizable, and scalable physical damage models tailored for extensive urban structures but applied at the building level. A key aspect of the solution is the development of generic building models that reflect typical characteristics related to structure, materials, and construction methods, thus eliminating the need for specific data and calculations for each individual building. This facilitates a more efficient assessment of flood impacts on urban environments. Vulnerability is assessed for a representative building drawn of a portfolio of structures with similar characteristics. The urban mixed-use pattern is categorized into various buildings classes, such as typical 19th/20th century masonry residential buildings or reinforced concrete administrative structures built after 1945 (Vetter et al. (2024)). Variability in geometry, material strength, and other relevant parameters within each class is incorporated into the analysis through probabilistic distributions (Neuhaeuser et al. (2024)).

The assessment framework comprises three sequential modules presented in figure 5. The first module utilizes open geodata to categorize buildings and develop geometric models, while literature research addresses data gaps. A parametric modeling approach is employed for simulation-based probabilistic analyses, starting with automated identification of building types and the generation of parametric base models which are then modified for structural analyses (Vetter et al. (2024)). The second module prepares structural models for probabilistic analyses and conducts the analyses (Neuhaeuser et al. (2024)). The module allows for various loading scenarios, including standard loads and those resulting from hazardous events, such as flooding, by varying flood height and flow velocity. Structural models representing specific building classes are derived by considering uncertainty and adjusting input parameters according to statistical distributions. The third module collects and evaluates results to derive fragility curves, describing the probability of reaching or exceeding certain damage levels based on intensity measures (Neuhaeuser et al. (2024)). All developed software tools are fully integrated, enabling automatic execution of modules for seamless integration into the co-simulation framework. This semi-automated procedure facilitates the development of fragility curves for assessing building vulnerability in urban contexts, as exemplified in Rosin et al. (2024).

To integrate into the co-simulation framework, georeferenced classifications of buildings and respective fragility curves for the building types are submitted. The latter are then overlaid with water levels within the co-simulation framework. As a result, the damage probability is determined spatially and event-dependently for each building type using the corresponding fragility curve.

3.1.7 Integration of situation data via spontaneous volunteers

The here-presented co-simulation requires external input data: the state of the simulated infrastructure at the beginning of the simulation serves as the initial conditions and further information at later times can be used to update and recalibrate the co-simulation. This recalibration is necessary since the co-simulation will naturally diverge from the real state of the infrastructure with increasing simulation time, reducing its predictive power. In particular, probabilistic and non-deterministic simulation components yield non-trivial result spaces of possible

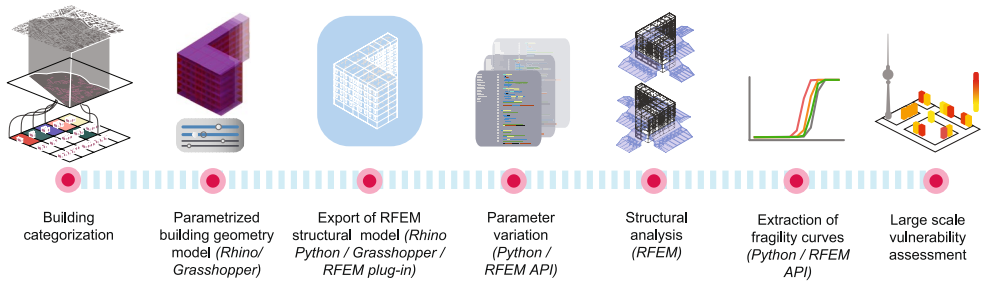


Fig. 5 Concept of building vulnerability assessment with specification of software (in italic): Rhino (Robert McNeel & Associates (2024)) and Grasshopper (Rutten (2024)) for geometric building modeling, RFEM (Dlubal Software (2024)) for structural modeling and analyses, Python for automated workflow and integration, figure adapted from Rosin et al. (2024)

simulation solutions. Additionally, initial conditions might not be exactly known, making assumptions necessary and adding further ambiguity.

Technical monitoring systems based on sensors and self-reporting might be insufficient to provide this input data in cases of large-scale, cascading infrastructure damages when infrastructure components, on which monitoring systems depend, are affected. Moreover, many critical infrastructures, such as most built structures and road networks, are not monitored by automated technical systems.

Spontaneous volunteers, i.e., volunteers that are not part of an organized disaster response organization, can not only help with physical labor but also by submitting situational information, increasing the situational awareness of disaster managers (Fathi and Fiedrich (2023)). Information crowdsourcing can yield fast, reliable, and low cost results as well as give access to previously unavailable information (Steed et al. (2019); Bayazidy-Hasanabad et al. (2021); Mirbabaie et al. (2016)).

Information crowdsourcing is arguably already applied routinely for supply infrastructure: consumer reports using phone calls, apps, or dedicated websites are a common way of detecting damages to supply infrastructure close to the consumers, as for example, in low-voltage electricity networks (Al-Kanj et al. (2023)). Crowdsourced information gathering has been, for example, studied for weather data (Barras et al. (2019); Kempf and Wetterdienst (2021); Kox et al. (2021)), various natural disaster events (Bayazidy-Hasanabad et al. (2021); Yuan and Liu (2018); Castanhari et al. (2016); Feichtinger et al. (2020); Dharmapuri Sridhar (2017)), supply and transport infrastructure (Mao et al. (2020); Sun et al. (2016); Rapousis et al. (2015); Shoemaker et al. (2021); Schnebele et al. (2014)), the structural stability of built structures (Tsai et al. (2020)), and even social and psychological issues of the population (Biegert et al. (2019)).

In this work, the usage of volunteer-submitted information as input data for the co-simulation is explored. The crowdsourcing is modeled after the setup of smartphone apps for volunteer coordination (e.g., Betke et al. (2024); Holzhüter et al. (2024)): spontaneous volunteers receive dedicated information retrieval tasks and submit situational information using the app. The volunteer-submitted information is collected, processed, optionally visualized, and incorporated into the co-simulation.

The volunteer module is triggered when a task for information retrieval for an area is received. This task is broken down into individual subtasks for each volunteer in the corresponding area. Data converter units map the reported information accordingly to be used as input for other downstream modules.

3.1.8 Data filtering, aggregation, and visualization

The simulation models presented above generate huge amounts of data, especially in complex simulation scenarios, where multiple critical infrastructures are interconnected. While this extensive output is rich in information, it can be overwhelming and hinder timely decision-making. To manage this complexity effectively, the data need to be broken down into manageable and relevant subsets. Through targeted data aggregation, data filtering and processing, and user-specific visualization, stakeholders can filter out key insights from the extensive data generated by simulations, enabling them to make informed decisions quickly and efficiently (Winter et al. (2024)).

In a complex co-simulation, all simulation modules produce valuable data that can be used for both external and internal purposes. The first step in managing these data is internal filtering, which is based on specific domain knowledge and determines what information should be shared. This process ensures that only relevant data contributes to the creation of a comprehensive digital twin that represents the complex simulation scenario. Naturally, the available data vary, as each simulation module produces its individual output formats. The use of a common data model, the OGC SensorThings API (Liang et al. (2021)), a well-established open-source data model, overcomes the hurdles of different data formats. A dedicated data collection unit, based on the FROST API Server (Hertweck et al. (2019)), is responsible for aggregating all available data. This unit also acts as a storage system that provides a complete representation of the simulated scenario. However, this extensive collection often contains excessive information that is not necessary for various use cases. Further processing is therefore essential to generate customized visualizations and enable targeted information transfer. This is achieved through various visualization tools that can be seamlessly integrated into the presented framework or used as complementary tools to ensure that users receive the most relevant insights for their specific needs.

The web-based information system ELD (Elektronische Lagedarstellung, engl. Electronic Situation Dashboard) is already available within this framework. It links a wide range of data to support decision-making and is specifically designed to assist staff in crisis management by providing the necessary tools for effective analysis and response. Based on open standards, the ELD enables easy integration of external data sources and facilitates the collection, management, and visualization of raw data, thereby increasing its versatility (Fraunhofer IOSB (2025, 2022)). Within the co-simulation, the ELD serves as a visualization tool, as one of its strengths is its support for data with spatial references for integration and visualization on maps. This capability ensures that users can effectively visualize complex

scenarios and make informed decisions during critical situations. Interfaces to the OGC SensorThings API and FROST are already implemented in the ELD, enabling easy integration and visualization of the simulation data. In addition, the data can be pre-processed to reduce its volume and create user-defined visualizations. Employing individual visual layers, the ELD provides easy customization, allowing users to overlay or hide layers to aggregate or filter information as needed. This functionality enables the display of specific datasets, such as public electricity grid information or comprehensive network insights, with customizable icons and colors to quickly highlight relevant areas. Additionally, new layers can be implemented quickly, allowing the visualized information to be broken down further into detail in almost real time, enhancing the users' ability to analyze complex scenarios effectively. Section 3.3 presents various uses of the ELD based on the application example. Each figure provides a comprehensive overview of the situation, highlighting the most critical information regarding failures in the supply infrastructure for crisis management teams. Users with different objectives, such as infrastructure operators, could easily customize these visualizations to focus on specific components of a single infrastructure, such as pipelines and nodes.

In addition to the ELD, further visualization tools can be easily integrated into the framework to enable user-specific visualization, either by direct integration or by developing specific interface modules. A DigLT-FROST interface module has already been developed for connecting the DigLT (Digitaler Lagetisch, engl. Digital Map Table) to the framework. This enables the simulation results to be displayed on the DigLT, a flexible software platform designed for collaborative visualization, including virtual reality capabilities, which allows multiple user to engage concurrently on the same scenario through different devices (van de Camp et al. (2020); Hoppe et al. (2021)). Plans are already in place to integrate further visualization tools and to make the data accessible to new user groups.

3.2 Module interaction interfaces

The processing units described within this section constitute a heterogeneous collection of self-contained tools based on engineering models, numerical and agent-based simulation, as well as data evaluation. As data transformers, each unit requires a specific well-defined input data format to be transformed into corresponding output data. The specifics of the input and output formats are highly tool-specific and include a significant amount of expert domain knowledge. In most cases, adaptation between the individual interfaces of the units is required to incorporate them into the co-simulation environment.

Within the presented framework, this is realized via bilateral data converter units mediating between the individual processing units. Figure 6 depicts the unit network employed for the urban digital twin as an FMC block diagram. The boxes (active elements) represent the simulator units (e.g., power grid, gas grid, or emergency response), while the arrows indicate the information flow through channels (circles) typified by data converters. A collector unit is employed to collect the simulation outcomes and provide them to the visualization and analysis dashboard.

For each connection in the unit network, adequate data conversion from the upstream unit to the corresponding downstream unit must be ensured by employing one or more suitable data conversion units. For instance, information pertaining to building damages must be processed and interpreted in terms of consequences that are interpretable, for e.g., the power grid simulator, which represents the operational state of active grid elements. Subsequently, information about, e.g., power flows must be translated for further processing by the emergency response module. This construction-kit approach facilitates flexible integration of diverse heterogeneous tools and even facilitates straightforward replacement solely requiring the adaption of data converters.

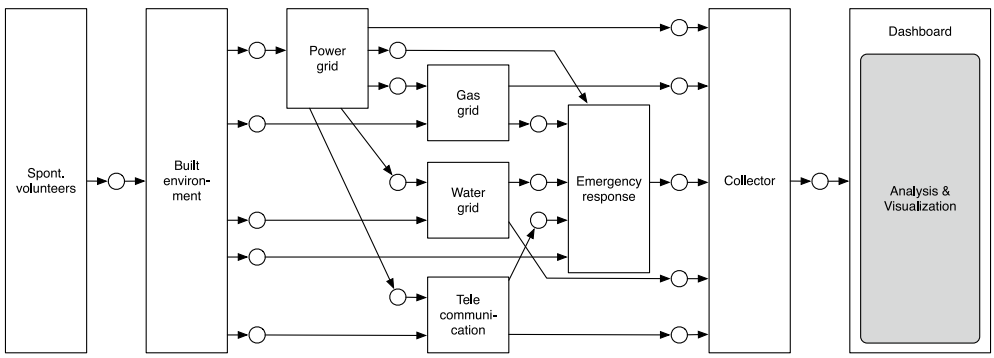
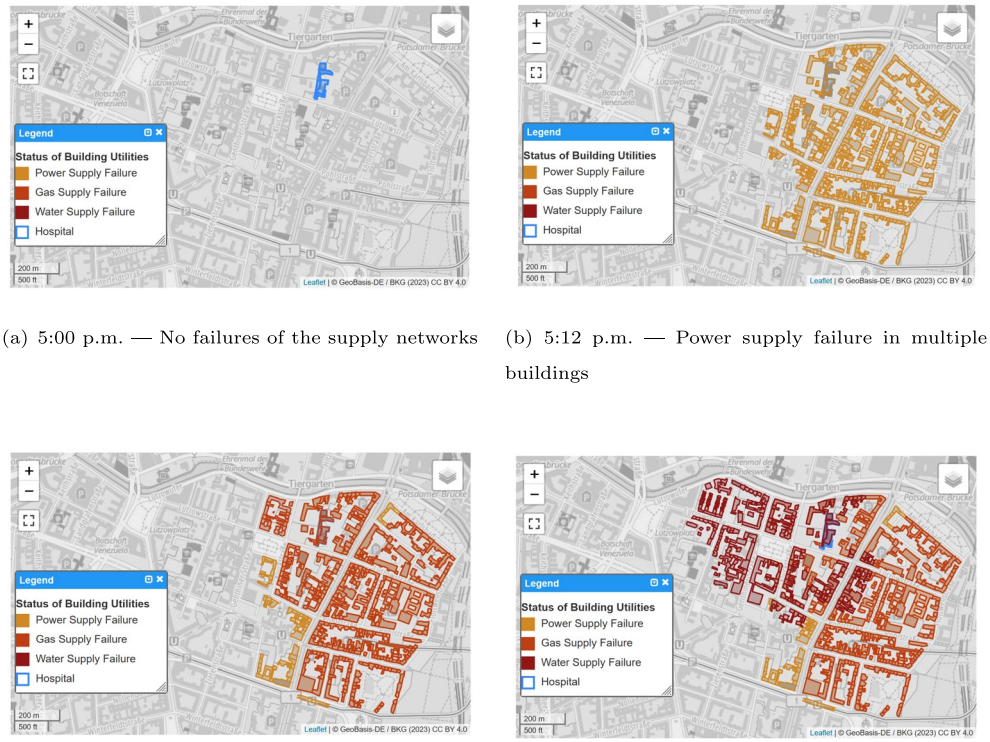


Fig. 6 Unit network depicting the module interaction considered in the example application with boxes representing active elements, arrows representing information flow through channels represented by circles

3.3 Example application: cascading failures of critical infrastructures in an urban environment

The functionality of the digital twin representation based on the simulation framework is illustrated using an exemplary scenario. The scenario takes place in an urban area of downtown Berlin (Germany), which is primarily characterized by residential buildings. Additionally, there are schools, an ambulance station, a hospital, and two substations in this area. The functionality of essential infrastructures in this urban area is simulated on October 20, 2024, from 5:00 p.m. to 5:19 p.m. with one simulation step each minute. At 5:10 p.m., a heavy rain event occurs. Due to the considerable amount of water and the fact that the urban area is sealed, the water drain is overloaded, the water accumulates, leading to initial flooding. Figure 7 illustrates the simulated area at four different points in time, each highlighting a different infrastructure. The buildings affected by supply infrastructure failures are indicated in color. The subsequently explained simulation is based on the assumption that the critical elements of the water, gas, and telecommunications grids lack emergency generators.

The input values used for the simulation include not only the start and end times but also the time steps and a map showing the water levels in the urban area. Individual emergencies are generated in the emergency response module up until 5:10 p.m.. The simulators for the supply infrastructures show regular operation without any failures during the period from 5:00 p.m. to 5:10 p.m. (see subfigure 7a). At 5:10 p.m., the predicted water levels resulting from the heavy rainfall event are transmitted to the spontaneous volunteers module which implements a simulation of information crowdsourcing of flood data shortly after the heavy rainfall event. The spontaneous volunteer module initially receives the volunteer configuration, i.e., the positions of the volunteers and additional information, a list of buildings, a map of the flooded area representing the real situation, and a list of polygons for report aggregation. The volunteer configuration of the example scenario comprises 250 simulated volunteers, 200 inside buildings, and 50 outside on sidewalks, with randomly chosen positions. Volunteers report the maximum flood depth visible to them. A rudimentary visibility simulation is implemented: volunteers can see approximately 50 m



(a) 5:00 p.m. — No failures of the supply networks

(b) 5:12 p.m. — Power supply failure in multiple buildings

(c) 5:15 p.m. — Gas supply failure in multiple buildings

(d) 5:17 p.m. — Water supply failure in multiple buildings

Fig. 7 Simulation of the impact of a heavy rainfall event on supply infrastructure in an urban area, illustrating four different timestamps that depict the cascading failures of various supply networks in specific parts of the area

and cannot see through buildings. Other visibility impairing features, e.g., trees or other objects, are neglected.

Volunteer-submitted flood reports are aggregated in quadratic polygons, which are 10,000 m^2 in size. The output of the spontaneous volunteer module is the averaged reported flood depth in each polygon. Figure 8 shows the resulting situational picture after the heavy rainfall event. A data converter unit maps the reported flood depth from the polygons to individual buildings by assigning each building in the respective polygon the corresponding flood depth. The water levels at individual buildings serve as the basis for calculating the probability of damage to the buildings. The analysis shows that some buildings affected by the flooding have a high probability of structural and/or secondary damage. In figure 8, buildings with a damage probability of more than 50% are shown.

In conjunction with the water levels, the emergency response module generates operations to pump out basements, for example, since many buildings get flooded. In addition, streets and bridges are impassable. Additionally, there are weather-unrelated incidents such as traffic accidents and medical emergencies. The latter incidents emerge within the emergency response simulation based on calibration data from the modeled area. These operations contribute to the continuous operational load (see figure 9a).

One of the substations in the area under consideration is also affected by the flooding, resulting in secondary damage to the building. Along with the high water level, the electrical systems in the substation fail, initially causing a power outage in part of the area (subfigure 7b). The other part of the area remains supplied by the second substation. Due to the power outage, the hospital in the area can no longer receive power from an external source, necessitating the activation of emergency generators to maintain supply. Overall, the failure in the power grid causes subsequent disruptions in the area. Wired communication comes to a complete standstill, and the base stations for mobile communications can no longer receive electricity. This results in limited means of communication for the population in the affected area. Additionally, gas compressors and gas valves can no longer be controlled, leading to a gas supply outage in parts of the area (subfigure 7c). The power supply to water pumps is also interrupted, resulting in a loss of water supply (subfigure 7d).

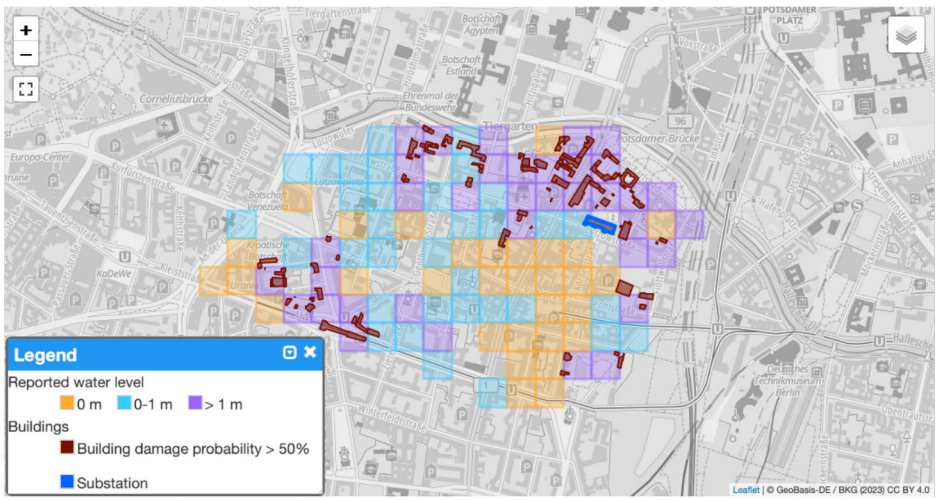
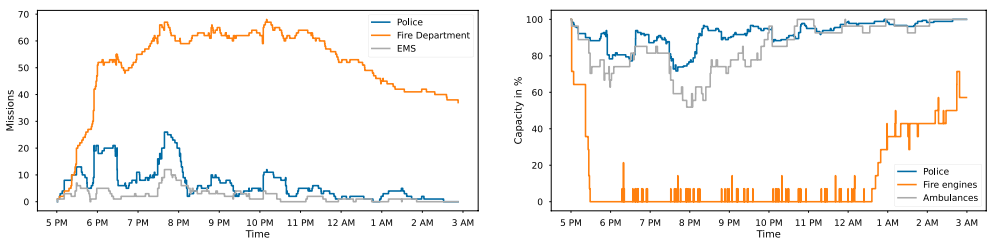


Fig. 8 Volunteer-reported water levels after the heavy rainfall event and buildings that have a damage probability of more than 50% as a result of the flooding



(a) Total number of missions (b) Available tactical units

Fig. 9 The total number of missions and the capacity (available tactical units) of the police, fire, and medical emergency services (EMS) over time in the simulated area

All outages generate additional operations for the emergency services to resolve the faults. However, the flooding makes it more difficult for the emergency services to reach the scene. In the simulated scenario, the hospital is impacted by interruptions to both gas and water supply. While emergency generators can compensate for power outages, there are no alternatives for the hospital's water and gas supply. As a result, the hospital qualifies for evacuation, necessitating a large-scale operation by emergency services. Figure 9b shows the remaining capacity of the police, fire, and medical emergency units. It can be seen that while the fire and rescue units are predicted to be at minimal capacity for most of the night, the medical emergency units will remain at a capacity that might allow for the additional load for a hospital evacuation in this area.

The results of the simulation reveal that heavy rainfall in urban areas can have significant impacts on various supply infrastructures due to the interconnectivity of the systems. The summarized failure damages, as shown in figure 10, indicate that a total of 445 households are without electricity, 289 households are without water, and 404 households are without gas. Furthermore, 84 buildings are severely affected by flooding and have a high probability of structural damages. It is important to note that individual buildings may also be impacted by multiple supply interruptions. In addition, the only hospital in the affected area would have to be evacuated due to failing water and gas supply, and the emergency services might reach their capacity limits to deal with all emergencies.

4 Conclusion and outlook

This article presents a simulation framework designed to assess and comprehend the interdependencies and cascading effects of disruptions among critical infrastructures in urban or sub-urban areas during disasters. The requirement to reproduce cascading effects within the simulation entails coupled rather than insular modeling of infrastructures. This requires defined ways of communication and translation methods between the modules. The proposed system architecture utilizes a microservice-based architecture. It facilitates communication and manages mutual triggering among simulation modules through event-based messaging and translation rules that connect the outputs and inputs of the microservices. Visualization tools like the web-based ELD or DigLT can be easily integrated to tailor the visualization of collected data on the status of infrastructure components to the needs of the use case.

The architectural concept presented not only serves the requirement of scalability, since new modules may be introduced in a straightforward manner by supplying suitable data converters implementing the required translation methods, but also serves the requirements in an environment where numerous developers and modules must collaborate in an interdisciplinary environment. This approach creates a foundation for bridging the gap in developing a digital twin representation including simulation tools enabling the analysis of interconnected critical infrastructure components within urban systems.

By design, the framework relies on specific scenario information and geographical references only as an input for the modular construction of the digital twin representation

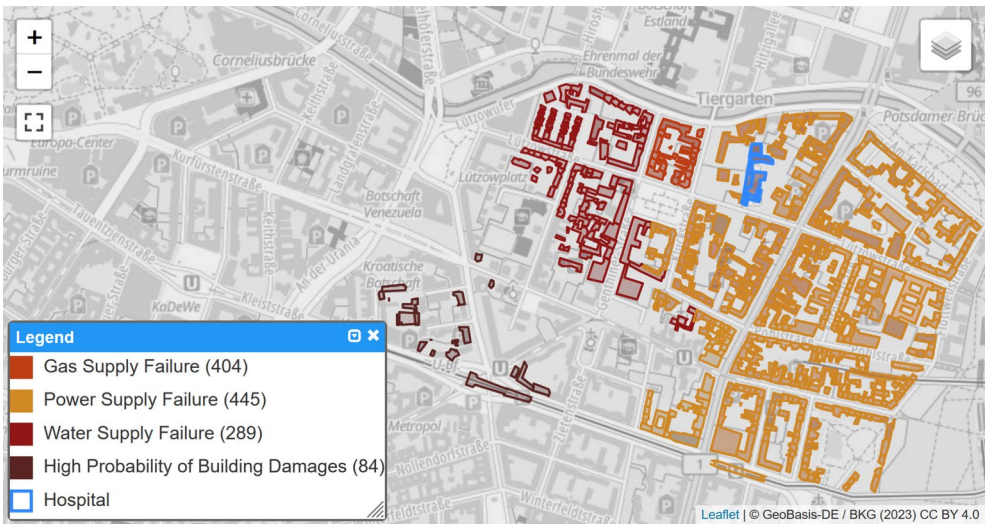


Fig. 10 Overview of all simulated outages and projected building damages. Individual buildings may be affected multiple times

which has to be provided by the user. Thereby, the underlying collection of simulator modules can be extended in a well-defined way to cover versatile areas of interests in "what-if" analyses. Thus, this generic approach is designed to be transferable to reproduce multifaceted scenarios in diverse urban environments in a straightforward manner, given that sufficient base data for adequate modeling are available. Utilization of this approach can produce valuable insights on the operational state of urban infrastructures and its impact on the general public before, during, and after disturbances.

The potential of the framework is showcased for an exemplary application scenario of cascading infrastructure failures in an urban environment in the aftermath of heavy rainfall. Several functional simulation modules, including those for gas, power supply, drinking water, telecommunication, building damage analysis, and emergency response, are incorporated as service units into the presented framework to facilitate their interplay via abstract interactions based on actual physical connections. In particular, all of the aforementioned simulation modules are based on generally valid, scientifically grounded models without the need for any data-driven, scenario-specific, or regional-specific training or special customization hindering transferability. This example application does not only demonstrate that the framework is able to account for inter-sectoral dependencies between connected infrastructures but also highlights the importance of such a simulation approach in order to uncover possibly unapparent vulnerabilities due to subtle, indirect interdependencies. It should be noted that this exemplary application only illustrates a small share of potential use cases for the presented methodology. Depending on ample data availability for realistic modeling, a plethora of heterogeneous simulation modules can be combined to best represent specific urban configurations due to the modular design of the framework. The individual simulation modules have to be connected via suitable bilateral data converter units mimicking mutual real-world interdependencies within the framework. Since these connections can be bidirectional, cascading disruptions might originate from various sources—within or even outside the modeled system. The entirety of all incorporated simulation modules, their respective base data, and interconnections together with exogenous triggers for disruptions constitutes the scenario definition to be provided by the user. However, the complexity of a scenario and thereby the amount of orchestration, i.e., inter-modular communication, might grow significantly when expanding its scope or refining its granularity due to extended combinatorics. Thus, incorporating additive hazard types or urban configurations different to those present in the example application might not only require more sophisticated dependency modeling by the user in terms of data converter units but might subsequently also have negative effects on simulation performance. These aspects have to be considered in regard to questions of scalability, adaptability, flexibility, and performance of the approach, as discussed in the following.

The presented research is to be understood as a proof-of-concept study to the newly developed approach and allows for further experimentation and development: The performance of the approach is initially demonstrated using an exemplary fictitious urban flooding scenario based on synthetic data, which indicates initial observed contingencies and interdependencies between the selected infrastructures. However, the capability of the coupled simulation to reliably model realistic socio-technical systems must be further validated against real-world data with a particular focus on interdependencies. This validation also must consider the data, parameters, and assumptions underlying the modeling of the real-world systems as well as the progress of particular hazard events since any digital twin representation can only be as good as its data basis and level of simplification. As a first step if not carried out already in case of existing, well-established tools, the performance of each individual simulation module has to be benchmarked and optimized. This can be achieved by comparing simulated values of system-state variables to real-world data measured for various physical system elements in normal-operation mode as well as to data collected during disruptions—if available. Secondly, validation of the newly introduced modeling of the interplay between the individual simulation modules realized through the novel coupled co-simulation framework is similarly important but possibly more challenging. Comprehensive research on or even first-time acquisition of cross-sectoral, time-resolved, suitable data, especially during relevant real-world hazard events with a focus on the monitoring of cascading effects is required for appropriate benchmarks. For assessing the impact of the accuracy of the simulations' predictions on hazard-event analyses, suitable metrics and key-performance indicators might be defined, potentially weighted according to vulnerability and importance rankings of the system elements.

The inclusion of additional simulation modules must be carried out to verify scalability and flexibility of the approach. Possible infrastructure components that can be integrated are, e.g., heat networks in energy systems and healthcare networks in the sector of critical infrastructure. The number and dimension of contingency scenarios that can be covered by this cross-domain coupled simulation framework also needs to be enhanced to provide versatility. Scenarios can be extended to technological, natural as well as man-made or hybrid attacks. Furthermore, relaxing the predefined scenario-based orchestration to enable spontaneous triggering of events to explore unforeseen contingencies and cascading effects of disruption may represent a future field for improvement.

Further improvement of the current setup could be achieved by incorporating more data instead of just flood depths and by adding more connections to other simulators. Weather information, information about more natural disaster events, and related damages are suitable candidates to be included in the co-simulation. These events could also trigger incidents in the emergency response simulation. Information about the usability of transport infrastructures, as streets, railways, bridges, tunnels, etc. could be included as well. Spontaneous volunteers can also report on the state of supply delivered by the infrastructures as well as on damages to infrastructure components, as for example damaged power lines.

A concept for more systematic incorporation of input data for the co-simulation has been proposed in Peitzsch et al. (2024). In this approach, the simulation is performed in discrete steps. Different possible initial state configurations are sampled using multiple simulation runs, for instance in a Monte-Carlo simulation approach. During the simulation, the fulfillment of boundary conditions is monitored. When new input becomes available, the simulation is stopped and the simulation results are checked for compatibility with input data. Incompatible simulation runs are discarded. An implementation and evaluation of this scheme remains for future work.

The framework can be tailored for conducting "what-if" analyses that emphasize the vulnerability and resilience of coupled socio-technical systems, training applications for relief units, task forces, or infrastructure managers. It can even be integrated into decision-support software utilized by decision makers, authorities, and organizations with security responsibilities. By combining simulation results with complementing real-world data obtained from, e.g., drone reconnaissance or spontaneous volunteers, at various stages of the crisis management cycle, authorities and infrastructure providers can effectively assess interdependencies and cascading effects of interconnected infrastructures.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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References

Al-Kanj L, Powell WB, Bouzaiene-Ayari B (2023) The information-collecting vehicle routing problem: Stochastic optimization for emergency storm response. arXiv preprint arXiv:2301.06497https://doi.org/10.48550/arXiv.2301.06497

Alvarez Lopez P, Behrisch M, Bieker-Walz L, Erdmann J, Flötteröd Y-P, Hilbrich R, Lücken L, Rummel J, Wagner P, Wießner E (2018) Microscopic traffic simulation using sumo. In: The 21st IEEE International Conference on Intelligent Transportation Systems. IEEE, https://elib.dlr.de/124092/

Andrae S, Pobuda P (2021) Agentenbasierte Modellierung. Springer, Wiesbaden. https://doi.org/10.1007/978-3-658-34953-0

Argonne National Laboratory: Repast Suite. Accessed: 2025-01-03 (2022). https://repast.github.io/

Barras H, Hering A, Martynov A, Noti P-A, Germann U, Martius O (2019) Experiences with >50,000 crowdsourced hail reports in switzerland. Bulletin of the American Meteorological Society 100(8):1429–1440. https://doi.org/10.1175/BAMS-D-18-0090.1

Bayazidy-Hasanabad M, Vayghan SS, Ghasemkhani N, Pradhan B, Alamri A (2021) Developing a volunteered geographic information-based system for rapidly estimating damage from natural disasters. Arab J Geosci 14:1783. https://doi.org/10.1007/s12517-021-08220-x

Biegert A, Kippnich U, Schneider L, Elkhazein S, Schulze M, Max M (2019) Web 2.0 und Soziale Medien Im Bevölkerungsschutz. Teil 2: Die Rolle Von Digital Volunteers Bei der Bewältigung Von Krisen- und Katastrophenlagen. DRK-Service GmbH

Baldo N, Miozzo M, Requena-Esteso M, Nin-Guerrero J (2011) An open source product-oriented lte network simulator based on ns-3. In: Proceedings of the 14th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems. MSWiM '11, pp. 293–298. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/2068897.2068948

Bandyopadhyay M, Singh V (2016) Development of agent based model for predicting emergency response time. Perspectives Sci 8:138–141. https://doi.org/10.1016/j.pisc.2016.04.017

Betke H, Sperling M, Schryen G, Sackmann S (2024) A design theory for spontaneous volunteer coordination systems in disaster response. 57th Annual Hawaii International Conference on System Sciences 2024. Proceedings

Chang K-T (2018) Introduction to Geographic Information Systems. McGraw-Hill Educiation

CloudEvents Authors: CloudEvents - A specification for describing event data in a common way. Accessed: 2025-01-24 (2024). https://cloudevents.io

Cabrera E, Luque E, Taboada M, Epelde F, Iglesias ML (2012) Abms optimization for emergency departments. In: Proceedings of the 2012 Winter Simulation Conference (WSC), pp. 1–12. https://doi.org/10.1109/WSC.2012.6465116

Castanhari RES, Rocha RdS, Andrade SCd, Albuquerque JPd (2016) A software architecture to integrate sensor data and volunteered geographic information for flood risk management. In: Proceedings of the ISCRAM 2016 Conference–Rio de Janeiro, Brazil, May 2016

Deshmukh A, Ho Oh, E., Hastak, M. (2011) Impact of flood damaged critical infrastructure on communities and industries. Built Environment Project and Asset Management 1(2):156–175. https://doi.org/10.1108/20441241111180415

Dlubal Software: RFEM 6. Accessed: 2025-01-28 (2024). https://www.dlubal.com/en/products/rfem-fea-software/rfem/what-is-rfem

Dharmapuri Sridhar MP (2017) Real-time flood mapping for disaster management decision support in chennai. PhD thesis, Massachusetts Institute of Technology

Feichtinger M, Wit R, Goldenits G, Kolejka T, Hollósi B, Žuvela-Aloise M, Feigl J (2020) Case-study of neighborhood-scale summertime urban air temperature for the city of vienna using crowdsourced data. Urban Climate 32:100597. https://doi.org/10.1016/j.uclim.2020.100597

Fathi R, Fiedrich F (2023) Digital volunteers in disaster management. In: Volunteered Geographic Information: Interpretation, Visualization and Social Context, pp. 265–276. Springer

Fraunhofer IOSB: Electronic Situation Dashboard for Civil Protection (ELD-BS). Accessed: 2025-01-27 (2022). https://www.iosb.fraunhofer.de/en/projects-and-products/electronic-situation-dashboard-civil-protection.html

Fraunhofer IOSB: WebGenesis© - System for networking a wide range of data to support decision-making. Accessed: 2025-01-27 (2025). https://www.iosb.fraunhofer.de/en/projects-and-products/webgenesis.html

Gütlein M, Djanatliev A (2020) Modeling and Simulation as a Service using Apache Kafka. In: Proceedings of the 10th International Conference on Simulation and Modeling Methodologies, Technologies and Applications, pp. 171–180. SCITEPRESS - Science and Technology Publications. https://doi.org/10.5220/0009780501710180

Grishin E, Gerasimov G, Gryazina E (2024) Overview of python power flow solvers. In: 2024 6th International Youth Conference on Radio Electronics, Electrical and Power Engineering (REEPE), pp. 1–6. https://doi.org/10.1109/REEPE60449.2024.10479730

Ganter S, Martini T, Kopustinskas V, Zalitis I, Vamanu B, Finger J, Dolgicers A, Zemite L, Fuggini C, Haering I, Stolz A (2023) A highly robust gas network simulation approach through an inherently solvable problem formulation for network states far from intended design point. Applied Mathematical Modelling 127:297–326 https://doi.org/10.1016/j.apm.2023.12.009

Giuseppe P, Nicola B, Marco M (2012) An lte module for the ns-3 network simulator. ACM. https://doi.org/10.4108/icst.simutools.2011.245571

Grainger JJ, Stephenson WD, Chang GW (2016) Power System Analysis, 2nd edn. McGraw-Hill Education, New York

Godfrey T, Zschaler S, Batra R, Douthwaite S, Edgeworth J, Edwards M, Miles S (2023) Supporting emergency department risk mitigation with a modular and reusable agent-based simulation infrastructure. In: 2023 Winter Simulation Conference (WSC), pp. 162–173. https://doi.org/10.1109/WSC60868.2023.10407894

Holzhüter M, Betke H, Fröhlich A, Klafft M (2024) Improving emergency and disaster response management through bidirectional communication between citizens and decision-makers–a field test on the coordination of spontaneous volunteers. ISCRAM Proceedings 21

Hayes S, Dent C, Mawdsley B, , Judson R, Collingwood T (2022) CReDo: an Overview. https://doi.org/10.17863/CAM.82908. https://www.repository.cam.ac.uk/handle/1810/335547

Hahn S, Rose DM, Sulak J, Kurner T (2015) Impact of realistic pedestrian mobility modelling in the context of mobile network simulation scenarios. In: 2015 IEEE 81st Vehicular Technology Conference (VTC Spring), pp. 1–5. https://doi.org/10.1109/VTCSpring.2015.7145870

Hoppe A, Camp F, Stiefelhagen R (2021) Shisha: Enabling shared perspective with face-to-face collaboration using redirected avatars in virtual reality. In: Proceedings of the 4th ACM Conference on Human-Computer Interaction, CSCW3, vol. 4, pp. 1–22. ACM. https://doi.org/10.1145/3432950

Hertweck P, Schaaf H, Blume J, Hellmund T, Moßgraber J (2019) Management of sensor data with open standards. In: Proceedings of the International ISCRAM Conference, vol. 2019-May, pp. 1126–1139. https://doi.org/10.1016/j.rcns.2023.07.004

IEEE: IEEE Standard for Modeling and Simulation (M &S) High Level Architecture (HLA) Framework and Rules. IEEE Std 1516-2010 (Revision of IEEE Std 1516-2000), 1–38 (2010) https://doi.org/10.1109/IEEESTD.2010.5553440. Accessed 2025-02-18

Ikuno JC, Wrulich M, Rupp M (2010) System level simulation of lte networks. In: 2010 IEEE 71st Vehicular Technology Conference, pp. 1–5. https://doi.org/10.1109/VETECS.2010.5494007

Jardosh AP, Belding-Royer EM, Almeroth KC, Suri S (2005) Real-world environment models for mobile network evaluation. IEEE J Selected Areas Commun 23(3):622–632. https://doi.org/10.1109/JSAC.2004.842561

Johnson DB (1999) Validation of wireless and mobile network models and simulation. In: DARPA/NIST Network Simulation Validation Workshop

Klise KA, Bynum M, Moriarty D, Murray R (2017) A software framework for assessing the resilience of drinking water systems to disasters with an example earthquake case study. Environ Model Softw 95:420–431. https://doi.org/10.1016/j.envsoft.2017.06.022

Knöpfel A, Gröne B, Tabeling P (2006) Fundamental Modeling Concepts: Effective Communication of IT Systems. Wiley, Hoboken

Kox T, Rust HW, Wentzel B, Gober M, Bottcher C, Lehmke J, Freundl E, Garschagen M (2021) Build and measure: students report weather impacts and collect weather data using self-built weather stations. Australasian J Disaster Trauma Studies 25(3):79–87

Koliou M, Lindt JW, McAllister TP, Ellingwood BR, Dillard M, Cutler H (2018) State of the research in community resilience: progress and challenges. Sustain Resilient Infrastruct 5(3):131–151. https://doi.org/10.1080/23789689.2017.1418547

Kempf H, Wetterdienst D (2021) Experience from large-scale crowdsourcing via weather apps. Australasian J Disaster Trauma Studies 25(3):87–93

Koch Z, Yuan M, Bristow E (2020) Emergency response after disaster strikes: Agent-based simulation of ambulances in new windsor, ny. Journal of Infrastructure Systems 26: 06020001 https://doi.org/10.1061/(ASCE)IS.1943-555X.0000565

Liang S, Khalafbeigi T, Schaaf H (2021) Ogc sensorthings api part 1: Sensing version 1.1. Technical report, OGC® Implementation Standard. https://docs.ogc.org/is/18-088/18-088.html

Lukau E, Schiller J, Meissen U (2023) Towards efficient post-blackout emergency communication based on citizens' smartphone state of charge. In: Radianti, J., Dokas, I., Lalone, N., Khazanchi, D. (eds.) Proceedings of ISCRAM 2023, 20th International Conference on Information Systems for Crisis Response and Management, pp. 76–87. https://doi.org/10.59297/NFTL3450

Li T, Zhao W, Baumanis C, Hall J, Machemehl R (2024) A python extension in sumo for simulating traffic incidents and emergency service vehicles. In: Gupta, R., Sun, M., Brzev, S., Alam, M.S., Ng, K.T.W., Li, J., El Damatty, A., Lim, C. (eds.) Proceedings of

the Canadian Society of Civil Engineering Annual Conference 2022, pp. 527–539. Springer, Cham

Martínez Alzamora F, Lerma N, Ayala HJ, Vegas O. Niño (2019) Upgrade of the gisred application for the free analysis of wdn under gis environment. In: 17th International Computing & Control for the Water Industry Conference, CCWI 2019, Exeter, United Kingdom

MATSim Contributors & MATSim Association: MATSim. Accessed: 2025-01-03 (2025). https://github.com/matsim-org/matsim-libs

Mahdiraji SA, Holmgren J, Alshaban A, Mihailescu R-C, Petersson J, Fatah JA (2022) A framework for constructing discrete event simulation models for emergency medical service policy analysis. Procedia Comput Sci 210:133–140. https://doi.org/10.1016/j.procs.2022.10.129

Modelica Association Accessed: 2025-06-05 (2025). https://fmi-standard.org

Martini T, Rosin J, Vetter JZ, Neuhäuser S, Lukau E, Catal F, Boigk M, Simon M, Monteforte M, Gerold M, Phung W, Dietze S, Finger J, Brausewetter P, Nicolai S (2024) Towards a modular co-simulation framework for the assessment of cascading effects among critical infrastructures and the impact on citizens. In: Kopustinskas, V., Foretic, H., Asensio Bermejo, I. (eds.) Resilience Assessment: Methodological Challenges and Applications to Critical Infrastructures, pp. 110–120. Publications Office of the European Union, Luxembourg. https://doi.org/10.2760/2808748

Mirbabaie M, Stieglitz S, Volkeri S (2016) Volunteered geographic information and its implications for disaster management. In: 2016 49th Hawaii International Conference on System Sciences (HICSS), pp. 207–216. https://doi.org/10.1109/HICSS.2016.33. IEEE

Mao H, Thakur G, Sparks K, Sanyal J, Bhaduri B (2020) Mapping near-real-time power outages from social media. In: Social Sensing and Big Data Computing for Disaster Management, pp. 88–102. Routledge,

Murtaza Sara S, Thota RC, Uddin MYS, Bani-Yaghoub M, Sutkin G, Abourraja MN (2024) Patient flow modeling and simulation to study hai incidence in an emergency department. Smart Health 32:100467. https://doi.org/10.1016/j.smhl.2024.100467

Neuhaeuser S, Vetter JZ, Rosin J, Stolz A (2024) A computational simulation framework for probabilistic vulnerability analysis of the built environment. In: Proceedings of the 31st International Workshop on Intelligent Computing in Engineering, EG-ICE 2024, pp. 534–543

Peitzsch S, Böhmer D, Gerold M (2024) Recalibration of digital twins via mobile agents. In: Goti, A., Guillén, A., Chiachio, J., Chiachio, M. (eds.) Proceedings of the 64th ESReDA Seminar and Doctoral Workshop "Digital Maintenance in the Digital Twin Era". University of Deusto, University of Deusto, Bilbao

Praks P, Kopustinskas V, Masera M (2015) Probabilistic modelling of security of supply in gas networks and evaluation of new infrastructure. Reliability Eng Syst Safety 144:254–264. https://doi.org/10.1016/j.ress.2015.08.005

PSC & OSGeo: PostGIS. Accessed: 2025-01-27 (2023). https://postgis.net/

Protzmann R, Schünemann B, Radusch I (2017) Simulation of convergent networks for intelligent transport systems with vsimrti. Networking Simulation for Intelligent Transportation Systems: High Mobile Wireless Nodes, 1–28 https://doi.org/10.1002/9781119407447.ch1

Restel H (2023) SimulationOps Towards a Simulation as-a-Service Platform for Resilient Societies Using a Cross-domain Data Mesh. In: Radianti, J., Dokas, I., Lalone, N., Khazanchi, D., Baharmand, H. (eds.) Proceedings of the 20th International ISCRAM Conference, pp. 575–585. University of Nebraska at Omaha. https://doi.org/10.59297/NUER9877

Riley GF, Henderson TR (2010) The ns-3 network simulator. In: Wehrle K, Guenes M, Gross J (eds.) Modeling and Tools for Network Simulation, pp. 15–34. Springer. https://doi.org/10.1007/978-3-642-12331-3_2

Siegfried R, Lloyd J, van den Berg T (2018) A New Reality: Modelling & Simulation as a Service - CSIAC. https://csiac.dtic.mil/articles/a-new-reality-modelling-simulation-as-a-service/ Accessed 2025-02-18

Rapousis N, Katsarakis M, Papadopouli M (2015) Qowater: A crowdsourcing approach for assessing the water quality. In: Proceedings of the 1st ACM International Workshop on Cyber-Physical Systems for Smart Water Networks, pp. 1–6. https://doi.org/10.1145/2738935.2738946

Robert McNeel & Associates: Rhino 8.7.24. Accessed: 2025-01-28 (2024). https://www.rhino3d.com/

Rupp M, Schwarz S, Taranetz M (2016) The Vienna LTE-Advanced Simulators - Up and Downlink. Link and System Level Simulation Springer. https://doi.org/10.1007/978-981-10-0617-3

Rutten D (2024) Grasshopper. Accessed: 2025-01-28. https://www.grasshopper3d.com/

Rosin J, Vetter JZ, Neuhaeuser S, Stolz A (2024) Building vulnerability assessment on an urban scale: A case study on heavy rainfall events. E3S Web of Conferences 579: 03002 https://doi.org/10.1051/e3sconf/202457903002

Rossman L, Woo H, Tryby M, Shang F, Janke R, Haxton T (2020) Epanet 2.2 user manual. Technical Report EPA/600/R-20/133, U.S. Environmental Protection Agency, Washington, DC, U.S.A

Schnebele E, Cervone G, Waters N (2014) Road assessment after flood events using non-authoritative data. Nat Hazards Earth Syst Sci 14(4):1007–1015. https://doi.org/10.5194/nhess-14-1007-2014

Steed RJ, Fuenzalida A, Bossu R, Bondár I, Heinloo A, Dupont A, Saul J, Strollo A (2019) Crowdsourcing triggers rapid, reliable earthquake locations. Sci Adv 5(4):9824. https://doi.org/10.1126/sciadv.aau9824

Schrab K, Neubauer M, Protzmann R, Radusch I, Manganiaris S, Lytrivis P, Amditis AJ (2023) Modeling an its management solution for mixed highway traffic with eclipse mosaic. IEEE Trans Intell Transport Syst 24(6):6575–6585. https://doi.org/10.1109/TITS.2022.3204174

Shoemaker E, Randolph H, Bryce J, Narman HS (2021) Designing crowdsourcing software to inform municipalities about infrastructure condition. In: 2021 IEEE 18th International Conference on Smart Communities: Improving Quality of Life Using ICT, IoT and AI (HONET), pp. 38–43. https://doi.org/10.1109/HONET53078.2021.9615483. IEEE

Sun H, Wang Z, Wang J, Huang Z, Carrington N, Liao J (2016) Data-driven power outage detection by social sensors. IEEE Trans Smart Grid 7(5):2516–2524. https://doi.org/10.1109/TSG.2016.2546181

Tsai M-H, Chan H-Y, Liu L-Y (2020) Conversation-based school building inspection support system. Appl Sci 10(11):3739. https://doi.org/10.3390/app10113739

The AnyLogic Company: AnyLogic. Accessed: 2025-01-03 (2025). https://www.anylogic.de/

Thule C, Lausdahl K, Gomes C, Meisl G, Larsen PG (2019) Maestro: the into-cps co-simulation framework. Simulation Modelling Practice Theory 92:45–61. https://doi.org/10.1016/j.simpat.2018.12.005

Thurner L, Scheidler A, Schafer F, Menke JH, Dollichon J, Meier F, Meinecke S, Braun M (2018) pandapower - an open source python tool for convenient modeling, analysis and optimization of electric power systems. IEEE Transactions on Power Systems. https://doi.org/10.1109/TPWRS.2018.2829021

Lindt JW, Kruse J, Cox DT, Gardoni P, Lee JS, Padgett J, McAllister TP, Barbosa A, Cutler H, Zandt SV, Rosenheim N, Navarro CM,

Sutley E, Hamideh S (2023) The interdependent networked community resilience modeling environment (IN-CORE). Resilient Cities and Structures 2(2):123505. https://doi.org/10.1016/j.rcns.2023.07.004

van de Camp F, Hoppe A, Unmueßig G, Peinsipp-Byma E (2020) Cooperative and location-independent terrain assessment for deployment planning using a 3d mixed reality environment. In: Proceedings of the SPIE Conference on Virtual, Augmented, and Mixed Reality (XR) Technology for Multi-Domain Operations, vol. 1142604. https://doi.org/10.1117/12.2558593

Vetter JZ, Neuhaeuser S, Rosin J, Stolz A (2024) A categorization and parametric modeling approach using open geodata enabling building vulnerability assessment. ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci. X(4/W5):309–316 https://doi.org/10.5194/isprs-annals-X-4-W5-2024-309-2024

Virdis A, Stea G, Nardini G (2014) Simulte-a modular system-level simulator for lte/lte-a networks based on omnet++. In: 2014 4th International Conference on Simulation and Modeling Methodologies, Technologies And Applications (SIMULTECH), pp. 59–70. IEEE. https://doi.org/10.5220/0005040000590070

Wilensky U (1999) NetLogo. Accessed: 2025-01-03. https://ccl.northwestern.edu/netlogo/

Winter N, Monteforte M, Müller F, Rohrbach I, Hertweck P, Martini T (2024) User-tailored visualization of simulation and sensor data for efficient crisis management: Wip paper – it solutions for crisis management. In: Penkert B, Hellingrath B, Rode M, Widera A, Middelhoff M, Boersma K, Kalthöner M (eds.) Embracing the Crisis Management Lifecycle: 21st International ISCRAM Conference, Univeristy of Münster, Germany. https://ojs.iscram.org/index.php/Proceedings/article/view/43/56

Wrulich M, Weiler W, Rupp M (2008) Hsdpa performance in a mixed traffic network. In: VTC Spring 2008 - IEEE Vehicular Technology Conference, pp. 2056–2060. https://doi.org/10.1109/VETECS.2008.462

Wang Y, Yu J-Z, Baroud H (2022) Generating synthetic systems of interdependent critical infrastructure networks. IEEE Syst J 16(2):3191–3202. https://doi.org/10.1109/JSYST.2021.3126308

Yuan F, Liu R (2018) Feasibility study of using crowdsourcing to identify critical affected areas for rapid damage assessment: Hurricane Matthew case study. Int J Disaster Risk Reduction 28:758–767. https://doi.org/10.1016/j.ijdrr.2018.02.003